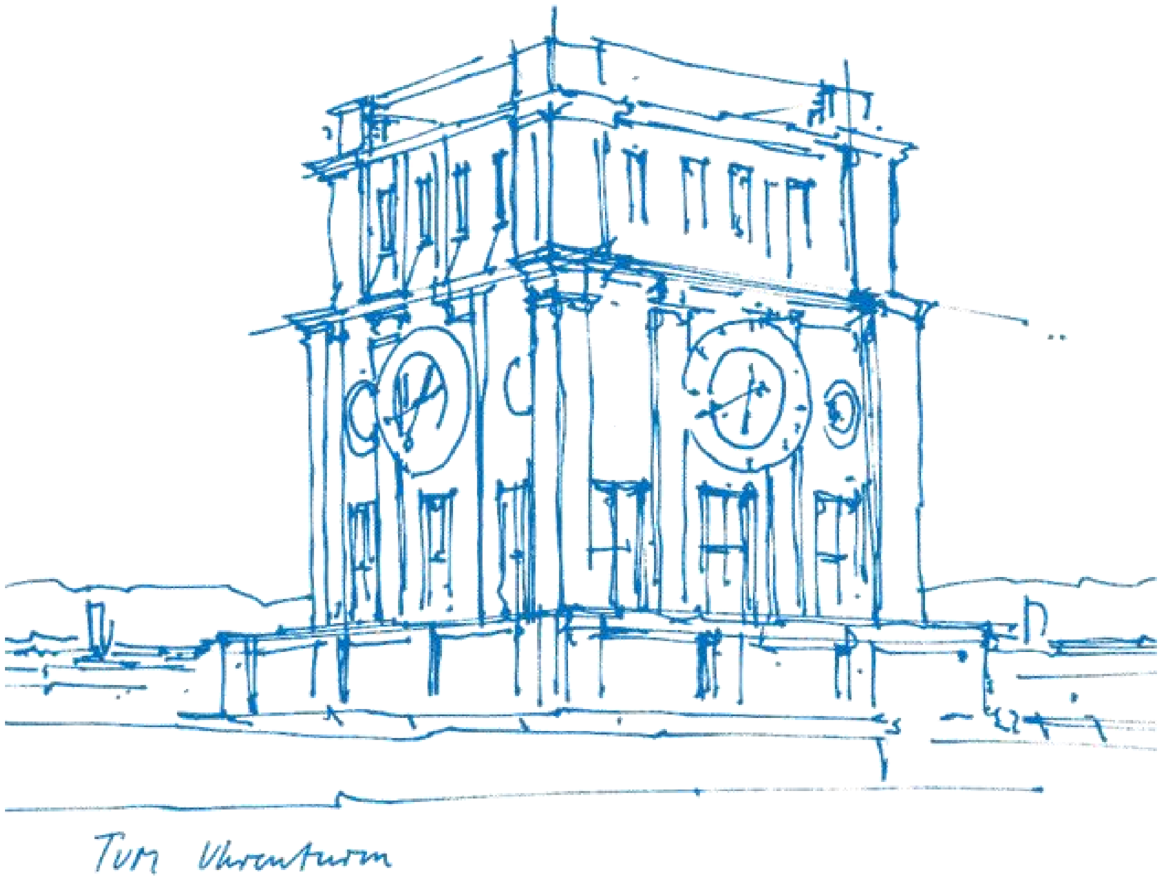
Discrete Probability Theory (IN0018)
School of Computation, Information, and Technology
Technical University of Munich

Tutorial 09

Central Limit Theorem (clt) and Convergence of RVs



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24. & 27. June 2025

About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week - 12:00-14:00 Tue, 03.11.018 (Group B33)
  - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercise (with solution)

https://drive.google.com/drive/folders1rBeL0NKHuYzrULK3Fr4CGr8svRhyTNOO?usp=sharing

I. Recap

- **(Review)**
  - ① - Markov's & Chebyshev's inequality
  - ② - Law of Large Numbers
    - - Weak law of large numbers (wlln)
    - - Strong law of large numbers (slln)

- **Central Limit Theorem (clt)**
  - ③ - Definition
  - ④ - Usage (with normal distribution table, see next page)

- **Convergence of Random Variables**
  - ⑤ - Definitions
    - - Convergence in Distribution
    - - Convergence in Probability
    - - Convergence in Mean
    - - Almost Sure Convergence

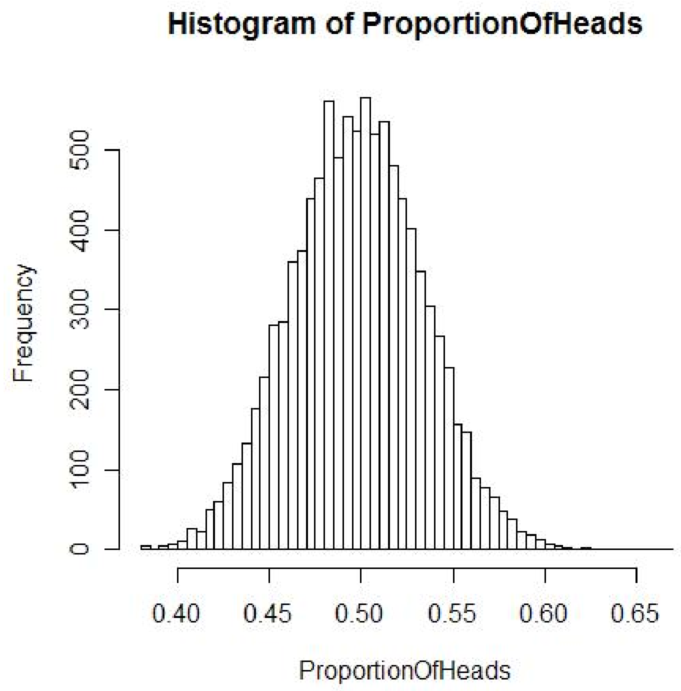

Image Source: https://en.wikipedia.org/wiki/Central_limit_theorem

I. Recap

Normal distribution table

e.g. $\Phi(1.56)=0.941$

$\Phi(2.82)=0.998$

$\Phi(x)$	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	0,500	0,504	0,508	0,512	0,516	0,520	0,524	0,528	0,532	0,536
0,1	0,540	0,544	0,548	0,552	0,556	0,560	0,564	0,567	0,571	0,575
0,2	0,579	0,583	0,587	0,591	0,595	0,599	0,603	0,606	0,610	0,614
0,3	0,618	0,622	0,626	0,629	0,633	0,637	0,641	0,644	0,648	0,652
0,4	0,655	0,659	0,663	0,666	0,670	0,674	0,677	0,681	0,684	0,688
0,5	0,691	0,695	0,698	0,702	0,705	0,709	0,712	0,716	0,719	0,722
0,6	0,726	0,729	0,732	0,736	0,739	0,742	0,745	0,749	0,752	0,755
0,7	0,758	0,761	0,764	0,767	0,770	0,773	0,776	0,779	0,782	0,785
0,8	0,788	0,791	0,794	0,797	0,800	0,802	0,805	0,808	0,811	0,813
0,9	0,816	0,819	0,821	0,824	0,826	0,829	0,831	0,834	0,836	0,839
1,0	0,841	0,844	0,846	0,848	0,851	0,853	0,855	0,858	0,860	0,862
1,1	0,864	0,867	0,869	0,871	0,873	0,875	0,877	0,879	0,881	0,883
1,2	0,885	0,887	0,889	0,891	0,893	0,894	0,896	0,898	0,900	0,901
1,3	0,903	0,905	0,907	0,908	0,910	0,911	0,913	0,915	0,916	0,918
1,4	0,919	0,921	0,922	0,924	0,925	0,926	0,928	0,929	0,931	0,932
1,5	0,933	0,934	0,936	0,937	0,938	0,939	0,941	0,942	0,943	0,944
1,6	0,945	0,946	0,947	0,948	0,949	0,951	0,952	0,953	0,954	0,954
1,7	0,955	0,956	0,957	0,958	0,959	0,960	0,961	0,962	0,962	0,963
1,8	0,964	0,965	0,966	0,966	0,967	0,968	0,969	0,969	0,970	0,971
1,9	0,971	0,972	0,973	0,973	0,974	0,974	0,975	0,976	0,976	0,977
2,0	0,977	0,978	0,978	0,979	0,979	0,980	0,980	0,981	0,981	0,982
2,1	0,982	0,983	0,983	0,983	0,984	0,984	0,985	0,985	0,985	0,986
2,2	0,986	0,986	0,987	0,987	0,987	0,988	0,988	0,988	0,989	0,989
2,3	0,989	0,990	0,990	0,990	0,990	0,991	0,991	0,991	0,991	0,992
2,4	0,992	0,992	0,992	0,992	0,993	0,993	0,993	0,993	0,993	0,994
2,5	0,994	0,994	0,994	0,994	0,994	0,995	0,995	0,995	0,995	0,995
2,6	0,995	0,995	0,996	0,996	0,996	0,996	0,996	0,996	0,996	0,996
2,7	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997
2,8	0,997	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998
2,9	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,999	0,999	0,999

Exercise 1 (heavy planes). A plane that can transport at most 100 passengers is filled with just enough jet fuel to carry up to 8 tons. For a fully booked flight, let X_i be a random variable modeling the i-th passengers weight in kg.

(i) Write down one (convincing) argument for why the $(X_i)_{i=1}^{100}$ should be modeled as independent or not as independent. From now on, assume they are iid with $\mathbb{E}[X_i] = 75$, and $\sigma[X_i] = 20$ for all $i \in [100]$.

(ii) Can you compute the exact probability that the maximum weight capacity of 8 tons is exceeded? Either compute it or explain why you can't?

(iii) Use the clt to compute an approximate probability that the maximum weight capacity will be exceeded.
[Hint: You can use wolframalpha or your favorite programming language/calculator to evaluate the cdf of standard Gaussians $\Phi(\cdot)$. Instead, you can also keep the results in terms of $\Phi(\cdot)$.]

(iv) If the profit margin per passenger is 3 Euros and the cost of going over capacity is 3000 Euros per flight (for topping up jet fuel), is the plane on average profitable for the airline (assuming it's always fully booked)?

II. Exercise Sheet

Exercise 2 (insurance claims and risk). An insurance company has $n = 10^4$ clients. For each client i, let X_i be the amount claimed in a year. Assume the claims $X_1, \ldots, X_n$ are iid RVRVs with mean $\mathbb{E}[X_i] = \mu = 500$ and variance $\mathrm{var}[X_i] = \sigma^2 = 800^2 = 640,000$. The company charges each client a premium of Z Euros.

(i) Let $S_n = \sum_{i=1}^{n} X_i$ be the total amount claimed. Calculate the minimum premium Z the company needs to charge to make an expected profit of half a million per year.

(ii) Use the weak law of large numbers to describe the behavior of the average claim per client $\bar{X}_n = S_n/n$ as n becomes very large.

(iii) The company wants to be at least 95% sure that the total premiums collected will cover the total claims S_n. Use Chebyshev's inequality to assess if the current number of clients $n = 10^4$ is sufficient with a premium of $Z = 550$.
[Hint: The company makes a loss if $S_n > n \cdot Z$. You want to bound $P(S_n > n \cdot Z)$.]

(iv) Now, use the clt to approximate the probability that the total claims S_n exceed the total premiums $n \cdot Z$. Compare this approximation with the bound obtained from Chebyshev's inequality. Why might the clt provide a different (and potentially more accurate) value for this probability? Which would you trust more?
[Hint: We need $P(S_n > n \cdot Z)$, which is easiest when S_n is standardized. You can use wolframalpha or your favorite programming language/calculator to evaluate the cdf of standard Gaussians.]

Exercise 3. Let X, Y RVRVs and $(X_n)_{n\in\mathbb{N}}, (Y_n)_{n\in\mathbb{N}}$ sequences of RVRVs. Verify the following statements.

(i) If $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$ then $X_n + Y_n \xrightarrow{P} X + Y$.
[Hint: Use the triangle inequality $|(X_n - X) + (Y_n - Y)| \leq |X_n - X| + |Y_n - Y|$. Decompose the probability to be bounded by $P(A \cup B) \leq P(A) + P(B)$ (this is called a *union bound* where A is an event only about X_n, X and B is only about Y_n, Y.]

(ii) If $\mathbb{E}[|X_n - X|] \to 0$, then $X_n \xrightarrow{P} X$.
[Hint: Use Markov's inequality for $|X_n - X|$.]

(iii) If $X_n \xrightarrow{P} X$ and $X_n \xrightarrow{P} Y$ then $P(X = Y) = 1$, i.e., X and Y are equal "almost surely."
[Hint: Again use the triangle inequality $|X - Y| \leq |X - X_n| + |X_n - Y|$, decompose $P(|X - Y| \geq \epsilon)$ via a union bound (for any n), to get to $P(|X - Y| \geq \epsilon) = 0$ for any $\epsilon > 0$ and use a limiting argument to get the statement also for $\epsilon = 0$ (for example via σ-subadditivity.]

(iv) If $\mathbb{E}[|X_n - X|] \to 0$ and $\mathbb{E}[|X_n - Y|] \to 0$ then $P(X = Y) = 1$.

II. Exercise Sheet

Exercise 4 (convergence of normal random variables). Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of RVRVs such that $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$ for each $n \in \mathbb{N}$.

(i) Show that if $\mu_n \to \mu \in \mathbb{R}$ and $\sigma_n^2 \to \sigma^2 > 0$ as $n \to \infty$, then $X_n \xrightarrow{d} X$, where $X \sim \mathcal{N}(\mu, \sigma^2)$.
[Hint: Use characteristic functions with the uniqueness and continuity properties. The cf of $\mathcal{N}(m, s^2)$ is $\phi(t) = \exp(imt - \frac{1}{2}s^2t^2)$.]

(ii) Show that if $\mu_n \to \mu \in \mathbb{R}$ and $\sigma_n^2 \to 0$ as $n \to \infty$, then $X_n \xrightarrow{P} \mu$.
[Hint: You can use either Chebyshev's inequality or the fact that convergence in distribution to a constant implies convergence in probability to that constant.]

III. Additional Exersice

Exercise 5. The student Avalon buys a new phone and a new laptop at the same time. According to the manufacturer, the lifespan X of the phone can be modeled as a normal distribution with an expected value of 5 (years) and a standard deviation of 3 (years). The lifespan Y of the laptop, on the other hand, is modeled as a normal distribution with an expected value of 7 years and a standard deviation of 4 (years). You may assume that X and Y are independent of each other.

(a) What is the probability that the difference between the failures of Avalon's phone and laptop is at most one year?
(b) The manufacturer offers an insurance policy for the phone. In this policy, a (euros) is paid to the customer if the phone fails before completing the first year. If a failure occurs during the second or third year, only $a/2$ (euros) is paid to the customer. Failures after the third year are not covered by the insurance. How must a be chosen so that the manufacturer expects to pay 50 euros per insurance policy?
Note: Since the range of a normally distributed random variable is always $\mathbb{R}$, the lifespans in this problem can also be negative. These do not need to be treated differently from positive lifespans.

III. Additional Exersice

1. (a) Aus Aufgabestellung gilt, dass $X \sim N(5,9)$, $Y \sim N(7,16)$.
Denn intuitiv sind X und Y unabhängig Normalverteilung,
nehm dann an, dass $Z = X - Y \sim N(-2, 25)$.
Gesamte Wahrkeit ist dann $Pr(|X-Y| \le 1) = Pr(|Z| \le 1) = Pr(-1 \le Z \le 1)$.
Angenommen dass $W = \frac{Z-\mu}{\sigma} = \frac{Z-(-2)}{5} = \frac{Z+2}{5}$.
$Pr(-1 \le Z \le 1) = Pr(\frac{-1+2}{5} \le W \le \frac{1+2}{5}) = \phi(\frac{3}{5}) - \phi(\frac{1}{5}) = 0.726 - 0.579 = 0.147$.

(b) Betrach nun nur Handy, also $X \sim N(5,9)$, angenommen $Y' = \frac{X-\mu}{\sigma} = \frac{X-5}{3}$.
Angenommen, dass A_1, A_2, A_3 für a, $\frac{a}{2}$, 0 Euro zahlen, A gemeinsam.
$Pr(A_1) = Pr(X \le 1) = Pr(Y' \le \frac{1-5}{3}) = 1 - \phi(\frac{4}{3}) = 1 - 0.908 = 0.092$.
$Pr(A_2) = Pr(1 \le X \le 3) = Pr(\frac{1-5}{3} \le Y' \le \frac{3-5}{3}) = \phi(\frac{4}{3}) - \phi(\frac{2}{3}) = 0.908 - 0.749 = 0.159$.
$Pr(A_3) = \phi(\frac{2}{3})$. $E(A) = a \cdot Pr(A_1) + \frac{a}{2} \cdot Pr(A_2) = 0.092a + 0.0795a = 50$, $a \approx 291.5$.